OFFICIAL TECHNICAL MANUAL

APM 2.8 Quadrotor Drone Complete Assembly & Configuration Guide

Integrated with FlySky FS-T6 Transmitter, FS-R6B 6-Channel PWM Receiver, and ArduCopter 3.2.1 on Mission Planner 1.3.68 (No-GPS Configuration).

FLIGHT CONTROLLER **APM 2.8 (Atmega 2560)**	RADIO SYSTEM **FlySky FS-T6 + FS-R6B**	FIRMWARE & GCS **ArduCopter 3.2.1 / MP 1.3.68**
AIRFRAME CONFIGURATION **Quad-X (450mm Frame)**	NAVIGATION SENSOR **Onboard Internal Compass (No GPS)**	ACTIVE FLIGHT MODES **Stabilize / AltHold / Land**

GUIDE CONTENTS

01. APM 2.8 Board Hardware Architecture & Critical Jumpers	Section 1
02. FlySky FS-T6 to FS-R6B Receiver Binding Procedure	Section 2
03. Complete Wiring: FS-R6B Receiver to APM 2.8 Inputs	Section 3
04. Quad-X Motor Layout, ESC Outputs & Propeller Rotation	Section 4
05. Mission Planner 1.3.68 Calibration & No-GPS Parameter Setup	Section 5
06. ESC All-At-Once Calibration & Pre-Flight Arming Safety	Section 6

Compiled for Amal Jyothi College of Engineering B.Tech Project • Anti-Vibration Avionics Stack

01 APM 2.8 Board Architecture & Jumper Setup

The ArduPilot Mega (APM 2.8) is an 8-bit AVR autopilot based on the ATmega2560 microprocessor with onboard MPU6000 6-DOF accelerometer/gyro and MS5611 high-precision barometric pressure sensor.



Figure 1.1: APM 2.8 Flight Controller Functional Board Topography & Pinout Configuration

1. The MAG Jumper (Internal Compass Enable)

Located immediately adjacent to the GPS and I2C sockets. When using **No External GPS/Compass** puck:

- **Must be SHORTED (Jumper Cap ON):** Connects the onboard HMC5883L/QMC5883 magnetometer to the MCU I2C bus.
- *If missing or left open, the autopilot will raise a* `Compass Not Healthy` *error.*

2. The JP1 Jumper (Power Rail Isolation)

The JP1 header connects or bridges the power planes of the `INPUTS` and `OUTPUTS` rails:

- **With Power Module (Recommended): REMOVE JP1**. This separates clean 5.3V CPU power from noisy ESC inductive spikes.
- **With ESC BEC only:** Place JP1 ON to route 5V from ESC to APM MCU.

02 FlySky FS-T6 to FS-R6B Binding & Radio Setup

The FlySky FS-T6 transmitter communicates using AFHDS (Automatic Frequency Hopping Digital System) 2.4GHz protocol. The FS-R6B receiver outputs independent PWM signals on 6 separate channels.



Figure 2.1: FlySky AFHDS Wireless Binding Architecture and Terminal Layout

Four-Step Receiver Binding Procedure:

1. **Insert Bind Plug:** Insert the black loop bind jumper into the **`BAT` / `B/VCC`** port on the FS-R6B receiver.
2. **Power Receiver:** Supply 5V power to any available channel (e.g. connect a 3-wire servo cable from APM Input 1). The red LED will **flash rapidly** (binding mode ready).
3. **Transmit Bind Signal:** On the FS-T6 transmitter, **hold down the circular BIND KEY** on the front faceplate while sliding the power switch to ON.
4. **Solid Red Confirmation:** The LED on the receiver will switch from rapid blinking to **SOLID RED**. Disconnect receiver power, remove the bind plug, and cycle power to confirm permanent pairing.

Channel Mapping Table (Standard Mode 2 / AETR)

Channel	Flight Axis	FS-T6 Physical Control	Pulse Range (µs)	ArduCopter Role
CH 1	Aileron (Roll)	Right Stick (Horizontal)	$1000 - 2000$	Rolls drone Left / Right
CH 2	Elevator (Pitch)	Right Stick (Vertical)	$1000 - 2000$	Pitches drone Forward / Aft
CH 3	Throttle	Left Stick (Vertical, Unsprung)	$1000 - 2000$	Direct motor thrust / Climb rate
CH 4	Rudder (Yaw)	Left Stick (Horizontal)	$1000 - 2000$	Rotates drone CW / CCW
CH 5	Auxiliary 1	Switch `SwB` (3-Position Toggle)	$1000 / 1500 / 2000$	**Primary Flight Mode Selector**
CH 6	Auxiliary 2	Knob `VrA` / Switch `SwA`	$1000 - 2000$	In-Flight Tuning / Camera Tilt

03 Receiver (FS-R6B) to APM 2.8 Physical Wiring

The APM 2.8 incorporates dedicated 3-pin headers for Inputs 1 through 8. Because the FS-R6B provides true discrete PWM pulses for each channel, connections are made directly with standard DuPont servo wires without signal converters.

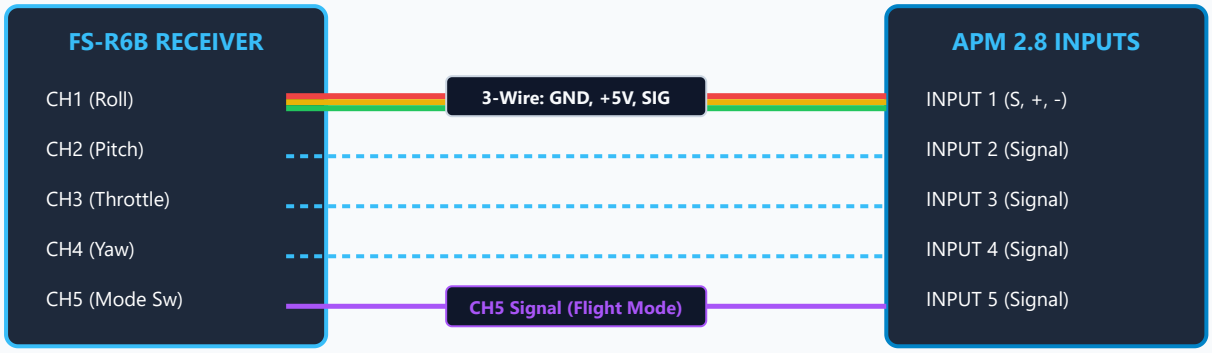


Figure 3.1: Complete Schematic Wiring Harness from FlySky FS-R6B to APM 2.8 Inputs

Three-Wire Power & Signal Rail (CH 1)

Connect a complete 3-conductor female-female servo wire between **CH1 on FS-R6B** and **INPUT 1 on APM 2.8**:

- **Inner Pin (APM):** White/Orange $\rightarrow$ Signal (PWM pulses)
- **Middle Pin (APM):** Red $\rightarrow$ $+5\text{V}$ Power Rail
- **Outer Pin (APM):** Black/Brown $\rightarrow$ System Ground

This single cable powers the internal radio receiver circuitry.

Single Signal Wires (CH 2 through CH 5)

To avoid ground loops and messy bundle clutter, connect only the **top Signal pin** for remaining channels:

- **CH 2 Signal** $\rightarrow$ APM **INPUT 2** (Elevator / Pitch)
- **CH 3 Signal** $\rightarrow$ APM **INPUT 3** (Throttle)
- **CH 4 Signal** $\rightarrow$ APM **INPUT 4** (Rudder / Yaw)
- **CH 5 Signal** $\rightarrow$ APM **INPUT 5** (Flight Mode Switch)

04 Quad-X Motor Layout, ESC Outputs & Propellers

ArduCopter uses the standardized **Quad-X** multirotor frame geometry. Motor order and rotational direction are strictly designated to balance aerodynamic gyroscopic torque.

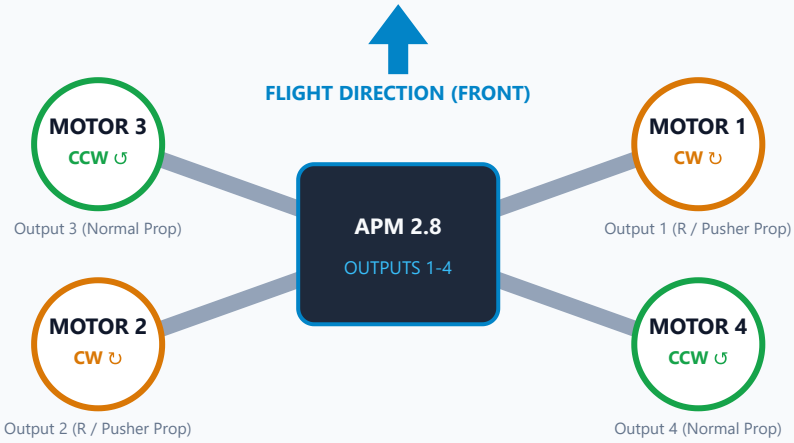


Figure 4.1: ArduCopter Quad-X Motor Layout, Directional Spin, and APM Output Mapping

Motor	Physical Frame Location	Spin Direction	Propeller Model	APM Output Rail Header
Motor 1	Front-Right Arm	**Clockwise (CW)**	1045R (Pusher / Reverse Pitch)	**OUTPUT 1** (Inner Signal, Mid 5V, Out GND)
Motor 2	Rear-Left Arm	**Clockwise (CW)**	1045R (Pusher / Reverse Pitch)	**OUTPUT 2** (Inner Signal, Mid 5V, Out GND)
Motor 3	Front-Left Arm	**Counter-Clockwise (CCW)**	1045 (Standard Direct Pitch)	**OUTPUT 3** (Inner Signal, Mid 5V, Out GND)
Motor 4	Rear-Right Arm	**Counter-Clockwise (CCW)**	1045 (Standard Direct Pitch)	**OUTPUT 4** (Inner Signal, Mid 5V, Out GND)

Motor Rotation Reversal Rule:

BLDC brushless motors spin in accordance with three-phase magnetic commutation. If a motor turns in the wrong direction during verification, **swap any two of the three motor wires** connected to that electronic speed controller (ESC). Never invert DC battery inputs!

05 Mission Planner 1.3.68 Configuration & No-GPS Tuning

Mission Planner 1.3.68 provides verified, glitch-free parameter flashing and sensor tuning for the legacy ATmega2560 APM processor.

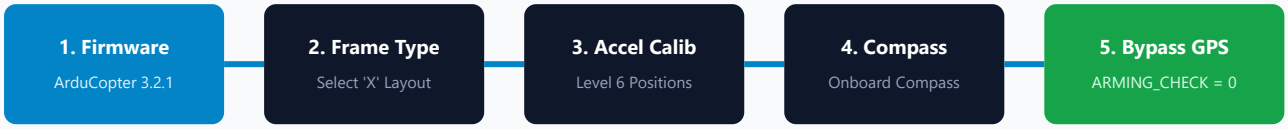


Figure 5.1: Sequential Calibration and Parameter Tuning Workflow in Mission Planner

1. Accelerometer 6-Axis Calibration

Go to `INITIAL SETUP` → `Mandatory Hardware` → `Accel Calibration`:

- Click **Calibrate Accel**.
- Place vehicle Level, Left, Right, Nose Down, Nose Up, and Back when requested by audio prompt.

2. Onboard Compass Live Calibration

Go to `Mandatory Hardware` → `Compass`:

- Select **APM with On-board Compass**.
- Orientation: **ROTATION_NONE**.
- Click **Live Calibration**, rotate vehicle 360° in all spheres until complete.

Critical No-GPS Arming & Failsafe Parameters

Because no external GPS unit is connected, ArduCopter will throw `PreArm: Need 3D Fix` and lock the arming sequence. You **MUST** adjust the following two parameters in `CONFIG/TUNING` → `Standard Parameters`:

1. **`ARMING_CHECK`**: Change from `All` to **`Skip GPS`** (or raw integer value `0` for bench testing).
2. **`FS_THR_ENABLE`**: Set to **`Enabled Always Land`** (or `Land`). By default, failsafe triggers `RTL` (Return to Launch), which causes an immediate flyaway or fatal crash without GPS coordinates!

06 All-At-Once ESC Calibration & Pre-Flight Checklist

Calibrating ESC throttle endpoints ensures all four motors commence rotation simultaneously and deliver identical linear thrust curves across the throttle band.

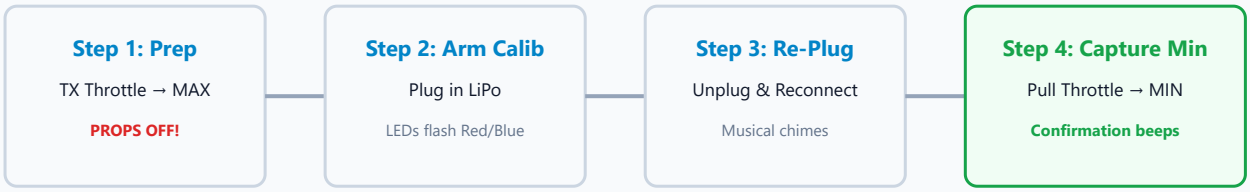


Figure 6.1: APM All-At-Once Simultaneous ESC Calibration Sequence

Pre-Flight Safety Readiness Checklist

- [ ] **Propellers Disconnected:** Keep props detached until full arming and motor rotation tests succeed.
- [ ] **Arming Sequence:** Turn on FS-T6, select `STABILIZE` mode, pull throttle stick down and hold **FULL RIGHT** for 5 seconds until the red APM arming LED turns solid.
- [ ] **Disarming Sequence:** Pull throttle stick down and hold **FULL LEFT** for 2 seconds.
- [ ] **Vibration Dampening:** Ensure APM 2.8 sits on silicone anti-vibration damping balls to prevent accelerometer saturation from frame vibrations.
- [ ] **Balance Check:** Center of Gravity (CoG) must reside precisely at the geometric intersection of the two diagonal frame arms.